

Topic 9.1

9.1: Comments on Reading Assignments for Module 8

Consider the dynamic int array example. As useful as it is, it's no help if you need a dynamic array of *float*. Wouldn't it be nice if there were a way you could reuse that code for dynamic arrays of any type? Templates give us that capability. Essentially, we are instructing the compiler to write code for us.

In developing a template, we first develop a concrete instance class, such as the dynamic int array. Then, we make the example generic. In the case of the dynamic array, the information we need to make generic is the type of data stored on the stack. We do this with template parameters.

When writing header files for a template class, the class declaration and the definition of all member functions go in the header file. This is because no code is generated when the compiler sees the templated function bodies. Code is only generated when you declare an object with a specific template argument.

Topic 9.2 (<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-9-dot-2>)